

# Task-Oriented Deep Joint Source–Channel Coding

Semantic communication for bandwidth-constrained image classification

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The project studies whether an edge device can send the information needed for a classification task without having to transmit and reconstruct every image detail.

Capstone First Review

Review window: 18–22 August 2026

Presentation slot: 15 minutes

Team and guide details to be added

# The project in one minute

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1

## Observe

An edge camera sees an image.

2

## Send

The link has limited bandwidth and added noise.

3

## Decide

A receiver predicts the image class.

4

## Compare

We measure task accuracy at the same channel budget.

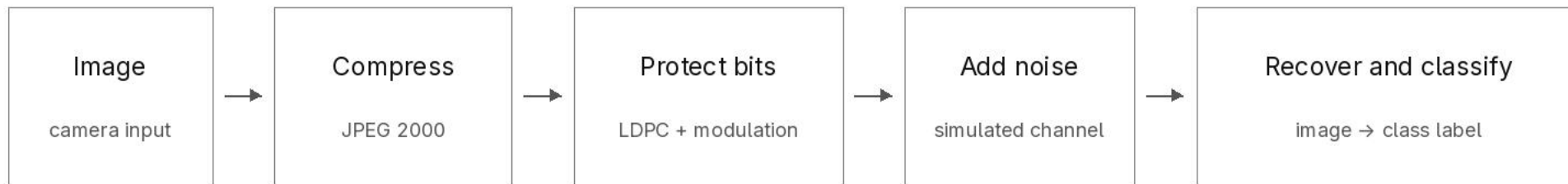
### Semantic communication

Send the information required to complete the task. Reconstructing every original pixel is optional when the task does not need it.

The final result is classification accuracy after transmission, not image quality by itself.

# How a conventional image link solves the task

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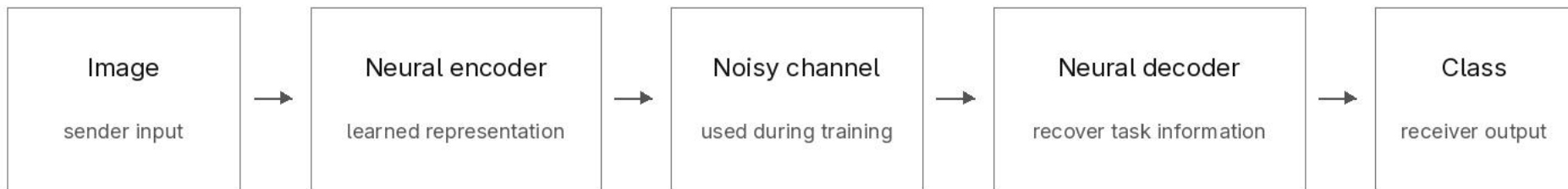


|                  |                                                              |
|------------------|--------------------------------------------------------------|
| Compression      | reduces the number of source bits.                           |
| Error correction | adds controlled redundancy so damaged bits can be recovered. |
| Modulation       | maps protected bits to symbols sent through the channel.     |
| Classifier       | uses the recovered image to choose a class label.            |

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Limitation: each stage is designed mainly to recover images and bits, while the final metric is whether the class decision is correct.

# What the learned joint system changes



## The central change

The sender and receiver are trained together through the channel. The transmitted representation can therefore preserve information useful for classification after noise.

## Why not send the label?

The project fixes the deployment split: the sender has an encoder, while the receiver owns the decoder and classifier. Sending a label would move the full task model to the sender and test a different system.

The method is supervised end-to-end learning. It is not reinforcement learning and it does not use an LLM.

# What prior research establishes—and leaves open

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Short-packet  
communication

Finite packets pay coding overhead and can fail abruptly.

Measure the practical baseline instead of assuming ideal capacity. [1–4]

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Learned image  
compression

Neural transforms can improve rate–distortion performance.

Good reconstruction quality is not the same as good task accuracy. [11–17]

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DeepJSCC

Learned continuous mappings can degrade gradually under noise.

Most image studies emphasize reconstruction rather than classification. [5–10,24]

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Task-oriented  
communication

Learned features can preserve inference under link limits.

Task awareness and joint coding are often not isolated from each other. [18–23]

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Gap addressed

A fair three-way comparison for image classification at equal channel use, with failures retained and one sealed test campaign.

# Problem statement and project objectives

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## Problem statement

Conventional image links protect reconstructed images and bits, while the receiver ultimately needs a correct class decision. A learned joint system may use the limited channel more effectively for that task, but the comparison must isolate task awareness from joint coding.

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## Research question

At the same bandwidth and channel conditions, can the learned joint system preserve classification accuracy better than a tuned conventional system, and is any remaining difference consistent with joint coding?

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## Objectives

1. Build the conventional image, digital feature and learned joint systems.
2. Match channel use, SNR, image identity and noise across all systems.
3. Freeze every model and operating choice before opening the test split.
4. Run one paired test campaign and report positive, null or adverse findings.

Completion depends on executing the fixed protocol correctly, not on obtaining a favorable result.

# Proposed methodology and comparison design

Shared input for every arm: same image · same dataset split · same complex-symbol budget · same SNR · same keyed noise

| System          | Transmission path                                                           | Scientific role                            |
|-----------------|-----------------------------------------------------------------------------|--------------------------------------------|
| Classical image | JPEG 2000 → framing → LDPC → modulation → AWGN → image → classifier         | Strong conventional reference              |
| Digital feature | Learned features → quantization → same digital physical layer → task head   | Separates task awareness from joint coding |
| Learned joint   | Neural encoder → power normalization → AWGN → neural decoder and class head | Tests end-to-end joint coding              |

## Evaluation controls

Tune on validation only; freeze before test access; retain failed transmissions in accuracy; count all bytes and physical-layer identities; compare paired per-image outcomes.

The test split remains sealed until gate G-12. No test result exists at First Review.

# Hypotheses and evaluation plan

## H1 — low-SNR separation (confirmatory primary)

Comparator: learned versus classical\_adaptive at the headline ratio. A grid point qualifies when the studentized paired mean  $\sqrt{N} \cdot \Delta(s) / \sigma(s)$  exceeds 1.96. Let  $R_{\text{obs}}$  be the longest run of consecutive qualifying points at or below the training SNR. H1 is supported only if  $R_{\text{obs}} \geq 3$  and the calibrated run p-value  $\leq 0.05$ .

The whole low-SNR-region mean paired difference is the effect size of record. A curve crossing is reported if observed but is not required.

H2 Does a fixed conventional configuration show a cliff while the learned system degrades more gradually?

H3 Does the learned–classical accuracy gap contract as SNR improves?

H4 Does the learned joint system also exceed the task-aware digital feature control?

Outcome rule: support, no support or an adverse result are all valid findings. The protocol does not change after test access.



# What has been demonstrated so far

| Gate | Observed evidence                                  | Plain-language meaning                                             | Boundary                                        |
|------|----------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------|
| G-1  | 89.8% validation accuracy                          | The reference image-classification task is viable.                 | Validation only; test is sealed.                |
| G-7  | 1.64 M parameters; measured GPU profile            | The planned learned model fits the available GPU.                  | A feasibility profile, not training.            |
| G-2  | Golden vectors and BLER conformance pass           | The digital physical-layer implementation behaves as expected.     | Small conformance evidence, not the full table. |
| W4   | Bounded conventional pipeline runs end to end      | Compression, coding, channel, decoding and records are integrated. | Integration evidence, not a comparison result.  |
| G8_C | C2 characterization paused at a durable checkpoint | Full-strength BLER measurement can resume safely.                  | Table, selection and training are unfinished.   |

No worker is running. No selection, training, validation decoding or test access has occurred.

# What remains after First Review

| Stage      | State            | Work                                                                               | Output               |
|------------|------------------|------------------------------------------------------------------------------------|----------------------|
| G8_C → G-8 | Paused → planned | Finish BLER characterization; freeze the table; select classical operating points. | Frozen baseline      |
| W5–W8      | Planned          | Train DJSCC and calibrate the task/reconstruction objective.                       | Frozen learned model |
| W9         | Planned          | Build and verify the task-aware digital feature control.                           | Attribution control  |
| W10–W12    | Planned          | Validation rehearsal; freeze all choices; run one sealed test campaign.            | Hypothesis decisions |
| W13–W17    | Planned          | Prepare demo, report, poster and final review package.                             | Final submission     |

First Review: 18–22 Aug · Second Review: 29 Sep–3 Oct · Final Review: 17–21 Nov · Report due: 20 Nov

W16 is report contingency, not room for new experimental scope.

# Application, scope and project risks

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## Real-world application

Remote cameras and edge sensors may need to support classification when bandwidth, latency or link reliability is limited. The receiver-owned task model represents split edge/cloud inference.

## Required project scope

Tier 1 is a simulation-based comparison of all three systems over normalized AWGN, with matched resources and one sealed test campaign. This is sufficient without radio hardware.

## Main risks and controls

BLER work is incomplete: resume from the authenticated checkpoint.

A hypothesis may not be supported: report the result without retuning.

Hardware adds confounders: keep it as gated stretch work.

**Technical claim boundary** OpenJPEG 2.5.4 and TS 38.212-derived LDPC/rate matching over abstract AWGN. The project does not claim a complete 5G NR radio link.

Outstanding human item: the guide's dated acknowledgement of the no-required-hardware path remains pending.

# Summary and next steps

## Project summary

|           |                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| Problem   | Remote classification must operate over a limited noisy link.                                                            |
| Proposal  | Train a task-aware joint encoder and decoder, then compare it with two matched digital systems.                          |
| Readiness | The reference task, model feasibility, digital physical layer and conventional pipeline have passed their current gates. |
| Next work | Finish characterization and selection, train the learned systems, freeze choices, then run the test campaign once.       |

## Six-criterion rubric map

|                   |              |
|-------------------|--------------|
| Motivation        | 2-4, 6, 11   |
| Objectives        | 6-7, 11-12   |
| Hypothesis        | 8            |
| Problem Survey    | 5            |
| Subject Knowledge | 3-5, 7-9, 11 |
| Time Plan         | 10           |

**First Review position**    The research question, three-system methodology, feasibility evidence and completion plan are defined. The final learned-versus-classical result remains future work.